# Inspection Intelligence that learns on the line

As electronics production becomes more complex and product mixes change faster, inspection systems need to adapt without adding programming and verification workload. Delvitech's AI-native architecture combines live neural inspection, self-programming and on-machine learning to support that process directly on the production line.

## Adapting inspection to production changes

New components, package variations and changing production conditions can require inspection models to be updated. In high-mix environments, the ability to handle this directly on the machine helps reduce setup time, specialist dependency and verification workload.

## Training Manager: AI that evolves with production

Training Manager lets customers train Delvitech's proprietary neural networks directly on the machine, using their own production images. Built for the process engineers and operators who run the line, it follows a three-step workflow: Spot, Teach, Reuse. The operator identifies a new component or production nuance, annotates the area of interest and fine-tunes the network. If a compliant solder joint is flagged because of lighting or a visual artefact, the operator reclassifies it and the network learns from the correction: fewer false calls, less verification time and a lighter repair-station workload. A proprietary continual learning technique adds new knowledge without overwriting what the network has already learned: learning without forgetting. A training history keeps track of past sessions and model versions.

## Inspection know-how that stays with the customer

Customers can introduce new components with their own teams, reducing their dependence on specialists and on individual operators. Training can be performed online or offline without stopping production: on offline programming stations, models are annotated, trained and validated in parallel with live operations, then deployed to the line. Validated models and libraries can be reused across products, machines, lines and sites, so what the team teaches the machine stays within the organisation as people and products change. The result is faster ramp-up on new products. Self-programming, previewed at productronica India 2026, is designed to speed up setup further by assisting component recognition and recipe creation.

## Horus: one AI-native architecture, multiple configurations

Delvitech is AI-native by design: proprietary neural networks are embedded across the inspection architecture from day one and recognise and classify board features during live inspection. Across the Horus portfolio, true-3D imaging, FPGA processing, Neith software and on-machine learning operate within the same integrated system.

Horus supports 3D SPI, pre-reflow AOI, post-reflow AOI and additional inspection applications on one configurable platform. The portfolio includes Horus, Horus Midi and Horus Max, with configurations for different board formats, throughput and production requirements.

Horus is designed for most PCBA and electromechanical inspection applications. Horus Midi extends the platform to wider boards and higher-throughput configurations, including Dual Lane. Horus Max is designed for larger and longer board applications, handling boards up to 815 × 715 mm and 15 kg.

Delvitech develops its inspection systems in Rancate, Switzerland, and supports customers in India from Bangalore, with a local demo centre and a 65,000 sq ft manufacturing facility expected to become operational in early 2027.

As production requirements evolve, Delvitech's AI-native architecture connects live inspection, programming and on-machine learning, while keeping inspection know-how within the customer's organisation.

## Inspection Intelligence, AI-native AOI and SPI.

DELVI TECH

HORUS

3D SPI, pre-reflow and post-reflow AOI on one platform.




Find out more at **www.delvi.tech**